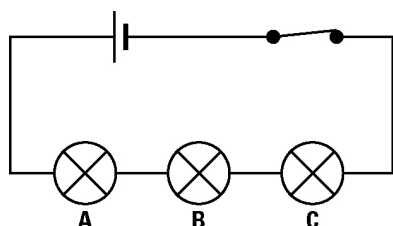


Series and parallel circuits

Student: Class:

Circuit one

1. Consider the following circuit containing identical lamps A, B and C in series.



- (a) Compare the brightness of the three lamps.

.....

- (b) Compare the currents that flow through lamp A and lamp C.

.....

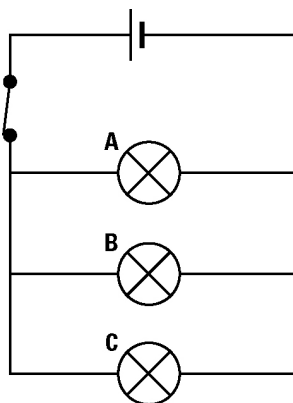
- (c) If the filament in bulb B burns out, will any of the lights remain lit? Explain.

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Circuit two

2. Consider the following circuit containing identical lamps A, B and C parallel. These lamps are the same as those for Circuit 1.



- (a) Compare the brightness of the three lamps.

.....

- (b) Compare the currents that flow through lamp A and lamp C.

.....

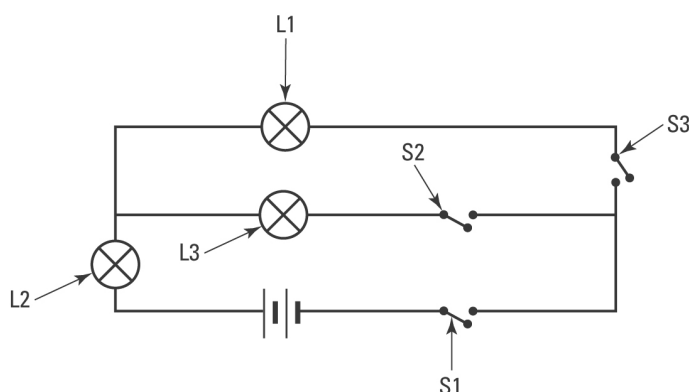
- (c) Compare the brightness of the lamps in Circuit 2 to their brightness in Circuit 1. Explain any differences.

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Circuit three

3. Consider the following circuit consisting of 3 lamps and 3 switches.



Predict which lamp(s) will light when:

- (a) switches S1 and S2 are closed

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- (b) switches S1 and S3 are closed

.....

- (c) switches S2 and S3 are closed.

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